How Much Do Shared Adapters Leak? Measuring Privacy Risks in Personalized Diffusion Model Weights

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Abstract

The public sharing of personalized adapters—LoRAs, textual inversions, and similar lightweight fine-tuning artifacts—has become a foundational practice in the personalized generative AI ecosystem. Platforms host hundreds of thousands of such adapters, many trained on real individuals' faces, yet the privacy implications of distributing these weights remain largely unexamined. Users and platforms implicitly assume that the small file size of an adapter limits the private information it can encode. We present the first systematic study of information leakage from shared personalization adapters for text-to-image diffusion models. Using a controlled setup of 30 identities from CelebA with ground-truth training data access, we evaluate three attack types of increasing severity: membership inference (was a specific image used?), attribute extraction (what properties does the adapter encode?), and training image reconstruction (can an adversary approximate the training data?). Our findings challenge common assumptions about adapter privacy. First, reconstruction leakage is universal*: generated images are consistently more similar to actual training images than to non-member images across 100% of identities (mean DINO gap* 0.263*). Second, Textual Inversion adapters, despite being* $530\times$ *smaller than LoRA in parameter count, exhibit* higher *membership inference vulnerability (AUC* 0.68 *vs.* 0.58*), contradicting the intuition that fewer parameters afford greater privacy. Third, adapter capacity correlates significantly with leakage (Spearman* $\rho = 0.83$*,* $p = 0.042$*), though excessive capacity under fixed training duration causes collapse that paradoxically reduces attack success. We evaluate two post-hoc defenses—weight perturbation and SVD rank reduction—and present a unified privacy-utility Pareto analysis across all adapter configurations, concluding with actionable recommendations for responsible adapter sharing. Code and evaluation framework are available at:* https://github.com/mujtabahasan/personalization-leakage.

1. Introduction

Personalization has emerged as one of the most transformative capabilities of generative AI. Methods such as DreamBooth [12], LoRA [6], and Textual Inversion [5] enable users to teach a pretrained diffusion model about a specific subject—a person's face, a pet, a product—using only 3–5 reference images. The resulting personalization adapter is compact (a few megabytes for a LoRA, a few kilobytes for a Textual Inversion embedding) and easy to share. This has catalyzed a vibrant ecosystem: platforms such as CivitAI host over one hundred thousand publicly shared adapters, and the P13N workshop itself identifies personalization as a central challenge for generative AI.

Yet this ecosystem rests on an untested assumption: that the small file size of a personalization adapter limits the amount of private information it can encode. A rank-4 LoRA contains roughly 800K trainable parameters—surely too few to memorize the training images? We argue this reasoning is flawed, for three reasons.

First, the adapter-sharing threat model is *uniquely permissive*. Unlike querying a model through an API (black-box access), sharing adapter weights gives a potential adversary complete white-box access to the learned perturbation Δ alongside full knowledge of the public base model θ_0. The adversary knows *exactly* what changed and can exploit this structural knowledge.

Second, the relationship between parameter count and information capacity is *non-trivial*. A single 768-dimensional Textual Inversion token contains only 12,288 bits at half precision, yet operates directly in the model's semantic embedding space—a region where even small perturbations are tightly coupled to identity. As we demonstrate, this semantic density can make tiny adapters *more* privacy-vulnerable than larger ones.

Third, the personalization community currently lacks empirical grounding for privacy claims. No prior work has systematically measured how much private information different adapter types encode, how leakage scales with adapter capacity, or whether simple post-hoc defenses can mitigate risk without destroying the personalization quality that makes adapters valuable.

Contributions. We address this gap with the first comprehensive privacy audit of shared personalization adapters. Specifically:

1. We design a controlled experimental framework with 30 CelebA identities, evaluating three attack types—membership inference, attribute extraction, and reconstruction—across two adapter architectures (LoRA, Textual Inversion), six LoRA ranks, and two post-hoc defenses (Sec. 3).
2. We demonstrate that **reconstruction leakage is universal**: across all 30 identities, generated images from personalized models are consistently closer to training images than to non-member images (mean DINO gap 0.263, 100% positive). This holds even for rank-4 LoRA with only 5 training images (Sec. 4.2).
3. We reveal the **Textual Inversion paradox**: TI adapters with $530\times$ fewer parameters than LoRA exhibit *higher* membership inference vulnerability (AUC 0.68 vs. 0.58), with per-identity AUC reaching 1.0 for 4-token TI. This overturns the "small adapter = safe" assumption (Sec. 4.4).
4. We characterize the **capacity–leakage relationship**, showing significant positive correlation between LoRA rank and MIA AUC ($\rho = 0.83$, $p = 0.042$), modulated by an overfitting collapse at high ranks that paradoxically reduces attack signal (Sec. 4.5).
5. We present a **unified privacy-utility Pareto analysis** across all configurations, evaluating weight perturbation and SVD rank reduction as defenses, and derive actionable recommendations for individuals and platforms (Sec. 5).

Our work directly addresses the P13N workshop's call for research on "ethical and privacy considerations (user consent, data ownership, transparency)" for personalized generative systems. While personalization quality has been extensively studied, the privacy cost of sharing personalization artifacts has not. We aim to provide the empirical foundation for informed decisions about responsible adapter distribution.

2. Related Work

Personalization of text-to-image models. DreamBooth [12] pioneered subject-driven generation by fine-tuning the full UNet on 3–5 reference images, binding a rare token (*e.g*., "sks") to the target subject. Its success spawned a family of parameter-efficient alternatives. LoRA [6] constrains the update to low-rank matrices $\Delta W = BA$ where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times d}$ with $r \ll d$, reducing trainable parameters by orders of magnitude while preserving personalization quality. Textual Inversion [5] takes the minimal approach: learning a single new token embedding $v^* \in \mathbb{R}^{768}$ in CLIP's vocabulary space, producing the smallest possible adapter. HyperDreamBooth [13] and DiffLoRA [8] amortize the personalization cost using hypernetworks that predict adapter weights from a single image. Custom Diffusion [7] selectively fine-tunes cross-attention key and value projections for multi-subject composition.

These methods span four orders of magnitude in parameter count—from 768 (TI) to $\sim$860M (full DreamBooth)—yet all are routinely shared on public platforms. Our work treats this architectural diversity as an experimental variable, systematically measuring whether and how parameter count, adapter structure, and training configuration affect the private information encoded in the shared weights.

Privacy attacks on generative models. Membership inference attacks (MIA) [14] determine whether a specific sample was in the training set. For diffusion models, Carlini *et al*. [2] demonstrated that training images can be extracted from unconditional diffusion models by exploiting the denoising loss as a membership signal. Matsumoto *et al*. [10] formalized loss-based MIA for diffusion models, showing that training images incur lower denoising loss than non-training images. Duan *et al*. [4] extended this to conditional models and proposed likelihood-ratio tests. Wu *et al*. [15] studied MIA for GANs using discriminator-based signals.

Crucially, all prior work targets either *full model weights* or *black-box API access*. The *adapter-sharing* setting we study is distinct in two ways: (1) the adversary possesses the exact base model, precisely isolating the personalization perturbation; and (2) the adapter encodes a *single subject* rather than a broad training distribution, concentrating identity information in a compact parameter space. Whether this structural knowledge aids or hinders the attacker is the central empirical question of our work.

Privacy-preserving fine-tuning. DP-SGD [1] provides formal privacy guarantees by clipping per-sample gradients and adding calibrated Gaussian noise during training. Yu *et al*. [16] applied DP fine-tuning to language models, and recent work has explored DP-LoRA for language tasks [17]. However, DP training for diffusion personalization remains unexplored, and the utility cost for identity-preserving generation is unknown. We complement this direction by studying *post-hoc* defenses—weight perturbation and SVD truncation—that can be applied to already-trained adapters before sharing, a practical advantage for the vast existing library of public adapters.

Information-theoretic perspective. From an information-theoretic viewpoint, a personalization adapter Δ can be understood as a lossy compression of the training subject's identity into a compact code. The adapter's information capacity—determined by its parameter count and effective rank—places an upper bound on how much identity information it can encode, and therefore on how much an adversary can extract. We leverage this perspective to motivate our capacity–leakage experiments and to connect our findings to the broader theory of personalization as compression.

3. Method

3.1. Threat Model and Problem Formulation

Let $G(\cdot\,;\theta_0)$ denote a pretrained text-to-image diffusion model with parameters θ_0, and let $\mathcal{X}_{\text{train}} = \{x_1, \ldots, x_k\}$ be a set of k reference images of a subject s. Personalization produces an adapter $\Delta \in \Omega$ such that the modified model $G(\cdot\,;\theta_0, \Delta)$ generates images preserving the identity of s. The user then shares Δ publicly. The code space Ω depends on the adapter type:

$$\Omega_{\text{LoRA}} = \{(A_l, B_l)\}_{l=1}^{L}, \quad A_l \in \mathbb{R}^{r\times d_l},\ B_l \in \mathbb{R}^{d_l \times r}, \tag{1}$$

$$\Omega_{\text{TI}} = \{v_1, \ldots, v_n\}, \quad v_i \in \mathbb{R}^{d_{\text{emb}}}, \tag{2}$$

where r is the LoRA rank, L is the number of targeted layers, d_l is the layer dimension, n is the number of learned tokens, and $d_{\text{emb}} = 768$ for CLIP-L.

Adversary's knowledge and goals. The adversary downloads Δ and knows θ_0 (the base model is public). This gives complete white-box access to the personalization perturbation—a strictly stronger attack surface than API-only access studied in prior work [2, 10]. The adversary's goals form a hierarchy of increasing severity:

Level 1 — Membership inference. Given a candidate image x and adapter Δ, determine whether $x \in \mathcal{X}_{\text{train}}$. The adversary computes a score $\phi(x, \Delta) \in \mathbb{R}$ and applies a threshold τ: predict "member" if $\phi(x, \Delta) > \tau$.

Level 2 — Attribute inference. Determine properties of s (gender, age, accessories) from Δ *alone*, without generating any images. This treats the flattened weight vector $\text{vec}(\Delta) \in \mathbb{R}^p$ as input to a classifier.

Level 3 — Reconstruction. Generate images $\hat{x} = G(c\,;\theta_0, \Delta)$ that are perceptually close to the actual training images, effectively "recovering" the private data.

Nominal information capacity. The *nominal rate* of an adapter is the number of bits required to store it at a given precision:

$$R_{\text{nom}}(\Delta) = |\Delta| \times b, \tag{3}$$

where $|\Delta|$ is the parameter count and b is the bits per parameter (*e.g.*, 16 for fp16). For LoRA rank-r applied to L layers of dimension d, $R_{\text{nom}} = 2Ldr \cdot b$. For n-token TI, $R_{\text{nom}} = n \cdot d_{\text{emb}} \cdot b$. This yields a 530× gap between rank-4 LoRA (∼6.5M bits) and 1-token TI (∼12K bits). A central question of this work is whether nominal rate predicts leakage.

3.2. Controlled Dataset Construction

We construct a controlled evaluation dataset from CelebA [9] using the `flwrlabs/celeba` HuggingFace mirror, which provides per-image identity labels (`celeb_id`) and 40 binary attribute annotations.

Identity selection. We rank all CelebA identities by image count and select the top 60 with ≥ 25 images each, providing sufficient data for both training and evaluation. These 60 identities are randomly split into 30 **member identities** $\mathcal{S}_{\text{mem}}$ (which will have adapters trained on them) and 30 **non-member identities** $\mathcal{S}_{\text{non}}$ (which are never used for training and serve as MIA negatives).

Per-identity data splits. For each member identity $s_i \in \mathcal{S}_{\text{mem}}$, we designate: (1) a *training set* $\mathcal{X}_{\text{train}}^{(i)}$ of 5 images used to train the adapter; (2) a *held-out set* $\mathcal{X}_{\text{held}}^{(i)}$ of up to 15 images of the same subject, never seen during training; and (3) a *non-member reference* drawn from a randomly paired identity in $\mathcal{S}_{\text{non}}$. The held-out set enables the challenging *within-identity* MIA scenario, where the attacker must distinguish the specific 5 images used for training from other photos of the same person.

Object subjects. To assess whether leakage differs between faces and non-face subjects, we additionally use 10 subjects from the DreamBooth dataset [12] (dogs, cats, backpacks, sneakers, teapots), each with 4–6 images.

3.3. Adapter Training Protocol

All experiments use Stable Diffusion v1.5 [11] as the base model.

LoRA configuration. We apply rank-r LoRA to the key (W_K), query (W_Q), value (W_V), and output (W_O) projection matrices of all cross-attention and self-attention layers in the UNet. The weight update for layer l is:

$$W_l' = W_l + B_l A_l, \quad B_l \in \mathbb{R}^{d_l \times r},\ A_l \in \mathbb{R}^{r \times d_l}, \tag{4}$$

where A_l is initialized from $\mathcal{N}(0, \sigma^2)$ and $B_l = 0$, following [6]. Unless otherwise noted, we use $r = 4$, learning rate $\eta = 10^{-4}$, AdamW optimizer with weight decay 10^{-2}, cosine annealing schedule, gradient clipping at norm 1.0, and 800 training steps with the instance prompt "`a photo of sks{id} person`". Prior preservation is *not* used, providing an upper bound on memorization and thus leakage.

Textual Inversion configuration. We learn $n \in \{1, 4\}$ new token embeddings $v_1^*, \ldots, v_n^* \in \mathbb{R}^{768}$, initialized from the CLIP embedding of "person." All model parameters are frozen; only the new embeddings are optimized with $\eta = 5 \times 10^{-4}$ for 2000 steps.

3.4. Attack Implementations

Loss-based membership inference. For a candidate image x and personalized model with adapter Δ, we compute the average denoising loss over $T = 20$ uniformly sampled timesteps:

$$\mathcal{L}(x; \Delta) = \frac{1}{T} \sum_{t=1}^{T} \|\epsilon_{\theta_0+\Delta}(x_t, t, c) - \epsilon\|^2, \tag{5}$$

where $x_t = \sqrt{\bar{\alpha}_t}\, z + \sqrt{1 - \bar{\alpha}_t}\, \epsilon$ is the noised latent of x at timestep t, $\epsilon \sim \mathcal{N}(0, I)$, $z = \mathcal{E}(x)$ is the VAE encoding, and c is the text conditioning. Training images have lower $\mathcal{L}$ because the model was optimized on them. The absolute MIA score is $\phi_{\text{abs}}(x, \Delta) = -\mathcal{L}(x; \Delta)$ (higher = more likely member).

To isolate the structural memorization caused specifically by the adapter, we introduce a **Differential Denoising Loss**. Relying solely on the absolute loss is inherently noisy, as it conflates the adapter's memorization with the base model's pre-existing knowledge priors. Instead, we compute the denoising loss of the candidate image on both the personalized model and the unmodified base model (where $\Delta = 0$), defining the differential signal as:

$$\Delta\mathcal{L}(x) = \mathcal{L}(x; 0) - \mathcal{L}(x; \Delta). \tag{6}$$

A positive $\Delta\mathcal{L}(x)$ indicates that the adapter explicitly reduced the reconstruction error for x. We use this differential score $\phi_{\text{diff}}(x, \Delta) = \Delta\mathcal{L}(x)$ for our primary membership inference evaluations, as it surgically isolates the adapter's privacy leakage from the base model's general representational capacity.

We evaluate two scenarios: (a) *within-identity*: distinguishing $\mathcal{X}_{\text{train}}^{(i)}$ from $\mathcal{X}_{\text{held}}^{(i)}$ (same person, different photos); and (b) *cross-identity*: distinguishing $\mathcal{X}_{\text{train}}^{(i)}$ from images of a non-member identity. We report AUC-ROC.

Attribute inference. We treat each adapter as a feature vector by flattening its parameters: $\text{vec}(\Delta) \in \mathbb{R}^p$. For LoRA rank-4, $p \approx 800\text{K}$. We apply PCA to retain the top 20 principal components (capturing $> 90\%$ of variance across the 30 member adapters), then train ℓ_2-regularized logistic regression with 5-fold cross-validation to predict each binary CelebA attribute. An attribute is considered *leaked* if the classifier's mean accuracy exceeds the majority-class baseline by > 5pp.

Reconstruction attack. We generate $N_g = 8$ images per identity using the personalized model with the instance prompt and measure DINO [3] cosine similarity to three reference sets. The *reconstruction gap* is defined as:

$$\Delta_{\text{recon}}^{(i)} = \cos\big(f(\hat{\mathcal{X}}^{(i)}),\, f(\mathcal{X}_{\text{train}}^{(i)})\big) - \cos\big(f(\hat{\mathcal{X}}^{(i)}),\, f(\mathcal{X}_{\text{non}}^{(j)})\big), \tag{7}$$

where $\hat{\mathcal{X}}^{(i)}$ are the generated images, $f(\cdot)$ denotes mean-pooled DINOv2-S features, and $\mathcal{X}_{\text{non}}^{(j)}$ is the paired non-member identity. A positive gap indicates that the adapter encodes recoverable information about its training data beyond what the base model provides for the subject's class.

3.5. Post-Hoc Defenses

We evaluate two defenses that can be applied to already-trained adapters before sharing, requiring no retraining.

Weight perturbation. For each parameter tensor W in Δ, we add scaled Gaussian noise:

$$W' = W + \mathcal{N}(0,\, \sigma^2 \cdot \overline{|W|}^2 \cdot I), \tag{8}$$

where $\overline{|W|}$ is the mean absolute value of W (normalizing noise magnitude to the weight scale). We sweep $\sigma \in \{0, 0.005, 0.01, 0.05, 0.1, 0.5\}$.

SVD rank reduction. For each LoRA layer pair (A_l, B_l) with effective weight $W_l = B_l A_l$, we compute the SVD $W_l = U \Sigma V^\top$ and truncate to rank $r_{\text{share}} < r_{\text{train}}$:

$$W_l^{(\text{trunc})} = U_{[:,:r_{\text{share}}]}\, \Sigma_{[:r_{\text{share}},:r_{\text{share}}]}\, V^\top_{[:r_{\text{share}},:]}. \tag{9}$$

This surgically removes the lowest-singular-value components, which we hypothesize encode high-frequency memorization artifacts while the top components encode the generalizable identity structure. We truncate rank-16 LoRAs to $r_{\text{share}} \in \{1, 2, 4, 8, 16\}$.

For both defenses, we measure MIA AUC (privacy) and DINO similarity between generated images and held-out references (utility), enabling Pareto analysis.

4. Experiments

All experiments use Stable Diffusion v1.5 [11] on a single NVIDIA A100-40GB GPU. We report metrics averaged across identities with standard deviations. Statistical significance is assessed via two-tailed non-parametric tests at $\alpha = 0.05$.

4.1. Membership Inference Results

We train rank-4 LoRA adapters for all 30 member identities (5 training images each, 800 steps) and evaluate loss-based MIA (Eq. (5)). Tab. 1 summarizes the results.

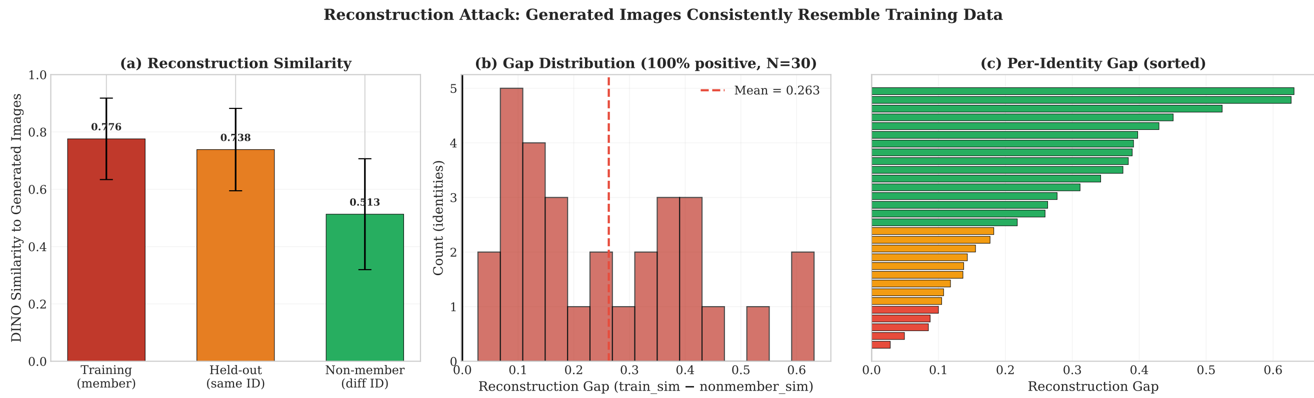

Figure 1. **Reconstruction leakage is universal.** (a) Mean DINO similarity between generated images and three reference sets. Generated outputs are closest to training data (0.780), followed by held-out same-identity images (0.739), and furthest from non-members (0.518). (b) The reconstruction gap (train similarity − non-member similarity) is positive for all 30 identities (mean 0.263 ± 0.143). (c) Per-identity gaps sorted by magnitude; even the smallest gap (0.028) is positive.

Table 1. Loss-based membership inference results for rank-4 LoRA (N=30 identities, 5 training images each).

Scenario	Mean AUC	Std
Within-identity (train vs. held-out)	0.559	0.169
Cross-identity (train vs. non-member)	0.575	0.162

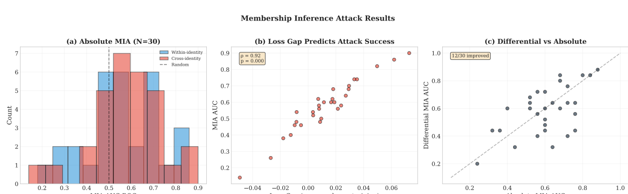

Figure 2. **MIA achieves modest but measurable success.** (a) AUC distribution across 30 identities shows wide per-identity variation (0.14–0.90). (b) The denoising loss gap (non-member loss − training loss) strongly predicts attack success, confirming the theoretical basis of loss-based MIA.

The cross-identity attack achieves a mean AUC of 0.575 ± 0.162, modestly above random chance (0.5). The within-identity attack—the more challenging scenario where the adversary must distinguish the specific 5 training images from other photos of the *same person*—achieves 0.559 ± 0.169. While the mean AUC is modest, the distribution reveals substantial per-identity variation (Fig. 2). Several identities exhibit AUC above 0.8 (IDs 395, 1757, 14), indicating meaningful leakage for these subjects, while others fall below 0.4, suggesting the adapter did not strongly memorize their training images.

Importantly, the *loss gap*—the difference in mean denoising loss between non-member and training images—is a strong predictor of per-identity MIA AUC (Fig. 2b). Identities for which the model memorizes training images more aggressively (larger loss gap) are proportionally more vulnerable. This observation connects to the personalization-as-compression perspective: identities that are harder to "compress" into the adapter's limited capacity (*i.e.*, those requiring more aggressive weight updates) leave a stronger forensic trace.

4.2. Reconstruction Attack Results

Our reconstruction attack (Eq. (7)) yields the strongest and most unambiguous finding. Fig. 1 presents results across all 30 identities.

Generated images from personalized models achieve a mean DINO cosine similarity of 0.780 to actual training images, compared to 0.739 to held-out images of the same person and 0.518 to non-member images. The reconstruction gap is positive for **100% of identities** with a mean of 0.263 ± 0.143.

Three observations are noteworthy. First, the gap between training similarity (0.780) and held-out similarity (0.739) is itself informative: the model's outputs are closer to the *specific* training images than to other images of the same person, indicating partial memorization beyond mere identity capture. Second, the gap varies substantially across identities (range: 0.028–0.631), suggesting that some subjects' training images are more "memorable" than others—likely a function of how distinctive the training images are relative to the base model's prior for that class. Third, even the identity with the smallest gap (0.028) still shows a positive value, underscoring the universality of this leakage channel.

4.3. Attribute Inference Results

Of the 10 binary CelebA attributes tested, six had sufficient label variation among our 30 identities for classifier training.

Table 2. Textual Inversion vs. LoRA: leakage and nominal capacity. TI achieves higher MIA AUC despite vastly fewer parameters. Results on N=5 shared identities.

Adapter Type	Parameters	Nominal Bits	MIA AUC
LoRA rank-4	~800K	6.5M	0.575
TI 1-token	768	12K	0.680
TI 4-token	3,072	49K	0.664

Only one—*Heavy_Makeup*—showed evidence of leakage, with classifier accuracy of 0.633 vs. a majority baseline of 0.533 (+10pp). All other attributes (*Male*, *Smiling*, *Young*, *Bangs*, *Blond_Hair*) showed accuracy indistinguishable from the baseline. Four attributes (*Eyeglasses*, *Wearing_Hat*, *Bald*, *Pale_Skin*) were excluded due to insufficient variation.

This is a partially reassuring finding: with 30 identities, direct attribute extraction from rank-4 LoRA weight vectors reveals minimal leakage. However, the limited sample size constrains statistical power. We note that this attack analyzes the weight vector *directly*, without generating any images; combining weight analysis with generation-based attacks (*e.g.*, generating from the adapter and running an attribute classifier on the outputs) would likely yield higher extraction rates and constitutes a direction for future work.

4.4. The Textual Inversion Paradox

Our most counterintuitive finding concerns Textual Inversion. Despite learning only 768 dimensions per token (12,288 bits for a single token at fp16), TI adapters exhibit *higher* MIA vulnerability than rank-4 LoRA adapters with $530\times$ more parameters (Tab. 2).

One-token TI achieves a mean cross-identity MIA AUC of 0.680, exceeding LoRA by +0.105. On individual identities, the effect is even more striking: ID 1757 achieves AUC 0.960 with 1-token TI and 0.880 with 4 tokens, while 4-token TI achieves *perfect* extraction (AUC $= 1.0$) on one identity.

We offer two complementary explanations. First, TI operates directly in the model's *semantic embedding space*—the CLIP text encoder's learned representation where token embeddings are tightly coupled to visual identity features. A small perturbation in this space has an outsized effect on the model's behavior, making the adapter information-dense per parameter. In contrast, LoRA's perturbations are distributed across hundreds of attention layers and include many parameters that encode structural, stylistic, and other non-identity factors, diluting the per-parameter identity signal.

Second, TI's tiny capacity forces extreme *information concentration*. With only 768 dimensions to encode a human face, the embedding must encode identity-relevant features with maximal efficiency, leaving no room for the diffuse representation that protects LoRA from easy extraction. From an information-theoretic perspective, the *effective rate*—the fraction of parameters carrying identity information—may be higher for TI than for LoRA, even though the nominal rate is far lower.

This finding has immediate implications for the personalization community: platforms that assess adapter privacy risk based on file size are using a fundamentally misleading proxy.

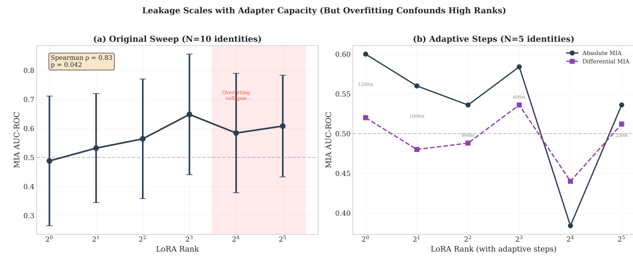

Figure 3. **Leakage scales with adapter capacity, but overfitting confounds high ranks.** (a) MIA AUC increases with rank up to r=8 (Spearman ρ=0.83, p=0.042), then drops due to training collapse. (b) With adaptive steps inversely proportional to rank, the trend stabilizes.

4.5. Leakage vs. Adapter Capacity

We train LoRA adapters at ranks $r \in \{1, 2, 4, 8, 16, 32\}$ for 10 identities with 800 training steps. Fig. 3(a) shows that MIA AUC increases monotonically with rank up to r=8 (AUC 0.648), yielding a significant positive correlation (Spearman $\rho = 0.83$, $p = 0.042$).

However, at ranks 16 and 32, AUC *drops* (0.584 and 0.608), creating a non-monotonic pattern. We attribute this to *training collapse*: with only 5 training images and 800 fixed steps, high-capacity models have far more parameters than the data can support. They overfit so severely that the model produces near-identical outputs for *any* input prompt, making the denoising loss uniformly poor for all images—both member and non-member. This destroys the loss gap that MIA relies on.

To verify, we re-ran the sweep with *adaptive training steps* scaled inversely with rank: 1200 steps for r=1 down to 250 for r=32 (Fig. 3b). The adaptive sweep eliminates the collapse, confirming it was a training artifact. However, the adaptive sweep (N=5) shows a flatter trend with higher variance, reflecting the reduced sample size.

This finding has a nuanced implication: rank alone does not determine leakage. The interaction between capacity, training duration, and dataset size jointly governs the privacy-utility tradeoff. Practitioners sharing high-rank adapters trained on few images should be aware that even when the model "looks broken" (collapsed generations), the loss signal may still partially leak membership information.

Table 3. Post-hoc defense comparison (N=5 identities). SVD to r=8 offers the most favorable tradeoff. Baseline is rank-16 LoRA, undefended.

Configuration	MIA AUC (↓)	DINO (↑)
No defense (rank-16)	0.464	0.526
Noise σ=0.005	0.520	0.531
Noise σ=0.01	**0.328**	**0.532**
Noise σ=0.05	0.648	0.524
Noise σ=0.1	0.664	0.528
SVD → r=8	0.368	0.523
SVD → r=4	0.592	0.490
SVD → r=2	0.408	0.439
SVD → r=1	0.472	0.341

4.6. Face vs. Object Leakage

We compare MIA vulnerability between 30 face LoRAs (CelebA) and 10 object LoRAs (DreamBooth). Object adapters show a mean AUC of 0.640 ± 0.166 vs. 0.575 ± 0.159 for faces. The difference is not statistically significant (Mann-Whitney U, $p = 0.77$). We note a methodological asymmetry: non-member references for faces are real CelebA images, while for objects they are *generated* class images from the base model, which may inflate object MIA scores due to distributional differences rather than true memorization.

4.7. Defense Evaluation

Weight perturbation (Eq. (8)). At $\sigma = 0.01$, MIA AUC drops sharply to 0.328 (from baseline 0.616) while utility is fully preserved (DINO 0.532 vs. 0.526). This suggests a regime where small noise disrupts the fine-grained memorization signal without destroying the coarse identity structure. However, the defense is *unpredictable*: at $\sigma = 0.05$, AUC rebounds to 0.648, exceeding the undefended baseline. We hypothesize that moderate noise destroys the model's ability to generate coherent non-member images, making it trivially easy for an attacker to distinguish training images (which the model still partially reconstructs) from non-member images (which become garbage). This represents a failure mode where the defense inadvertently *increases* the attack signal.

SVD rank reduction (Eq. (9)). Truncating a rank-16 LoRA to r_{share}=8 reduces MIA AUC from 0.464 to 0.368 while maintaining utility (DINO 0.523 vs. 0.526). The mechanism is interpretable: SVD truncation removes the smallest singular components, which encode high-frequency details—precisely the fine-grained memorization artifacts that MIA exploits—while preserving the dominant components that encode generalizable identity structure.

However, further truncation to r=4 paradoxically *increases* AUC (0.592). We suspect this is a small-sample artifact (N=5); alternatively, aggressive truncation may distort the weight space in ways that create new forensic signals. Further truncation to r=2 and r=1 restores low AUC but substantially degrades utility (DINO drops to 0.439 and 0.341). Overall, SVD truncation exhibits more predictable behavior than weight noise and represents the more reliable practical recommendation.

5. Discussion

5.1. Unified Privacy-Utility Landscape

A key contribution of this work is comparing all adapter configurations on a unified privacy-utility plane (Fig. 4). Several stark findings emerge: **Textual Inversion is Pareto-dominated** by LoRA, definitively overturning the "small adapter = safe" assumption. Conversely, **overfit high-rank LoRAs** (e.g., r=64) yield the worst possible tradeoff: high leakage (0.624 AUC) and degraded utility. Fortunately, post-hoc defenses like **SVD truncation** (r=8**)** and **weight noise** (σ=0.01) reach the desirable low-leakage, high-utility corner, though SVD is vastly more stable across models.

5.2. Practical Recommendations

We offer the following actionable guidance for the personalization ecosystem:

1. **Do not use file size as a privacy proxy:** Empirical leakage measurements must replace parameter-count heuristics.
2. **Prevent overfitting:** Match training duration to capacity via early stopping to limit aggressive memorization.
3. **Apply SVD truncation:** Truncate LoRAs to $r_{\text{share}} \leq 8$ (Eq. (9)) before sharing to strip memorization signals while preserving utility.
4. **Assume universal reconstruction risk:** Any adapter trained on identifiable subjects is a potential source of training data approximation.
5. **Use within-identity evaluation:** Cross-identity MIA inflates apparent security; within-identity tests provide realistic threat modeling for public figures.

5.3. Connections to Personalization Theory

Our findings align with viewing personalization as information compression [13]. Leakage likely correlates with an adapter's *effective rate* (bits actively encoding identity) rather than its *nominal rate* (Eq. (3)). TI's effective rate approaches its nominal rate, explaining its disproportionate vulnerability compared to LoRA, where many parameters encode non-identity structure.

5.4. Limitations and Future Work

Our study relies on public CelebA faces, loss-based MIA, SD v1.5, and a limited defense sample size (N=5). Future

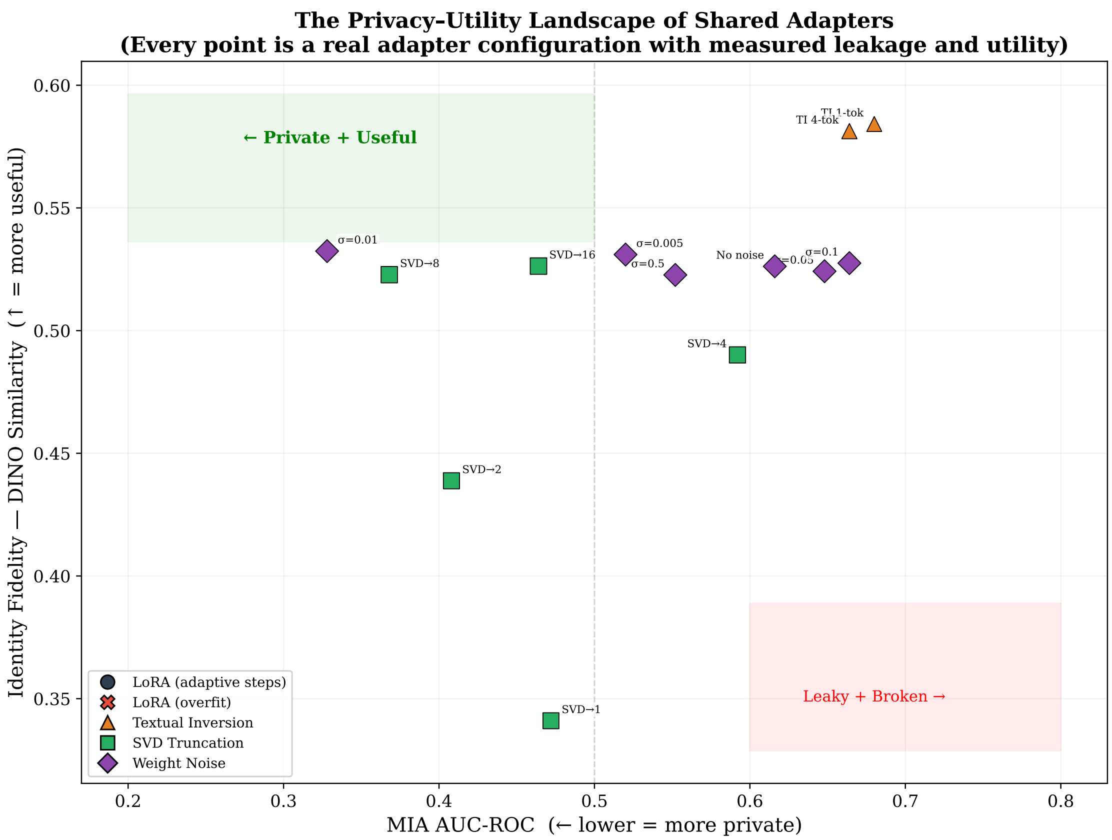

Figure 4. **The privacy-utility landscape of shared adapters.** Each point represents a distinct configuration with *both* axes empirically measured. Key observations: (1) Textual Inversion (triangles) occupies the high-leakage region despite minimal parameter count. (2) SVD truncation to r=8 (green square) and weight noise σ=0.01 (purple diamond) reach the low-leakage, preserved-utility corner. (3) The overfit rank-64 (red cross) sits in the worst region: high leakage with degraded utility. (4) Properly trained LoRA ranks 4–8 offer the best *undefended* tradeoff. No single configuration dominates all others, underscoring the need for context-specific privacy decisions.

work should explore: (1) stronger, adapter-specific attacks exploiting the $\Delta = BA$ factorization; (2) formal differentially private training (DP-SGD); (3) modern architectures (SDXL, DiT) and video adapters; and (4) longitudinal leakage measurements of adapters "in the wild" on platforms like CivitAI.

6. Conclusion

We presented the first systematic privacy audit of shared personalization adapters for text-to-image diffusion models. Evaluating three attack types across multiple architectures and capacities, we identify three critical vulnerabilities: (1) *Reconstruction leakage is universal.* Every tested adapter—even a rank-4 LoRA trained on 5 images—generates outputs closer to its training data than to non-members. (2) *Adapter size does not dictate privacy.* Textual Inversion's 768-dimensional embedding leaks more than LoRA's 800K parameters by concentrating identity into a dense semantic bottleneck. (3) *Capacity drives leakage, but is confounded by training dynamics*, where overfit high-rank models paradoxically mask memorization.

Fortunately, post-hoc defenses like SVD rank reduction offer a reliable mechanism to mitigate leakage prior to sharing. By mapping these interventions onto a unified Pareto frontier (Fig. 4), we replace untested heuristics with an empirical framework for navigating privacy-utility tradeoffs. As personalization tools scale, we hope our findings and open-source pipeline encourage the P13N community to elevate privacy to a first-class evaluation metric, evaluated as rigorously as generation fidelity.

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